# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

**Full Syllabus (Chapters 1–11)** | Class 7 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** acidic | Sour taste is a property of acids. |
| 2 | **(a)** an electric cell | A cell has chemicals that produce electric current. |
| 3 | **(b)** lustre | Metals reflect light. |
| 4 | **(b)** physical change | Only form or state changes. |
| 5 | **(d)** puberty | Puberty marks the beginning of reproductive maturity. |
| 6 | **(a)** lower temperature | Heat flows downhill in temperature. |
| 7 | **(a)** second | 1 minute = 60 seconds. |
| 8 | **(a)** ingestion | Ingestion is intake. |
| 9 | **(b)** autotrophs | Auto means self. |
| 10 | **(c)** straight lines | Rectilinear propagation. |
| 11 | **(b)** west to east | Counter-clockwise from above the North Pole. |
| 12 | **(d)** sulphuric acid | Sulphuric acid is a strong mineral acid. |
| 13 | **(c)** conductors | E.g. copper. |
| 14 | **(c)** brittle | They break when hammered. |
| 15 | **(a)** melting of wax | Wax solidifies again. |
| 16 | **(c)** puberty | Reproductive maturity. |
| 17 | **(d)** pass on heat to neighbours but stay in position | They vibrate. |
| 18 | **(d)** distance travelled per unit time | Speed = distance/time. |
| 19 | **(b)** villi | They increase surface area. |
| 20 | **(c)** blue-black | Starch test. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Substances that show the nature of a solution by colour change (1); Litmus and turmeric (1). | 1+1 |
| 22 | A complete path for the flow of current (1); It has a source, a switch and a device (1). | 1+1 |
| 23 | Lustre and malleability (1); Ductility and conductivity (1). | 1+1 |
| 24 | No new substance is formed (1); Melting of ice, tearing paper (1). | 1+1 |
| 25 | Period when body becomes sexually mature (1); About 10 to 12 years (1). | 1+1 |
| 26 | Conduction and convection (1); Radiation (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Dip blue and red litmus (1); Red from blue means acid; blue from red means base (1); No change means neutral (1). | 1+1+1 |
| 28 | The circuit is open (1); No current flows (1); The filament does not heat (1). | 1+1+1 |
| 29 | Make a circuit with a gap (1); Place metal and non-metal samples (1); Metals light the bulb; non-metals do not (except graphite) (1). | 1+1+1 |
| 30 | Melting: physical (1); Burning and rusting: chemical (1); Boiling: physical (1). | 1+1+1 |
| 31 | Rapid increase in height (1); Voice change (1); Growth of body hair (1). | 1+1+1 |
|  | **OR:** Fix pins with wax on a rod (1); Heat one end (1); Pins fall one by one (1). | 1+1+1 |
| 32 | T = 60/30 = 2 s (1); Oscillations in 1 minute = 30 (1); Time for one oscillation is 2 s (1). | 1+1+1 |
|  | **OR:** Mouth to oesophagus (1); Stomach to small intestine (1); Large intestine to anus (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Collect lemon juice, vinegar, soap, baking soda, salt (1); Test with litmus papers (1); Test with turmeric (1); Test with China rose (1); Record in a table and classify (1). | 1+1+1+1+1 |
| **34** | Make a circuit with a gap (1); Place objects in the gap (1); Copper wire, iron nail, coin: bulb glows (1); Rubber, plastic: no glow (1); Record in a table (1). | 1+1+1+1+1 |
| **35** | Lustre: gold (1); Malleability: aluminium (1); Ductility: copper (1); Conductivity: silver (1); Sonority: bell metal (1). | 1+1+1+1+1 |
| **36** | Physical: melting (1); Physical: tearing paper (1); Chemical: burning (1); Chemical: rusting (1); Chemical: souring of milk (1). | 1+1+1+1+1 |
| | **OR:** Growth spurt (1); Voice change in boys (1); Facial hair in boys (1); Breast development in girls (1); Menstruation in girls (1). | 1+1+1+1+1 |
| **37** | Conduction: spoon in tea (1); Convection: boiling water (1); Radiation: Sun (1); Conduction needs solid (1); Radiation needs no medium (1). | 1+1+1+1+1 |
| | **OR:** Hang pendulums of lengths 25 cm and 100 cm (1); Time 20 oscillations each (1); Find T = t/20 (1); Longer has greater period (1); Mass has no effect (1). | 1+1+1+1+1 |
| **38** | Mouth: chewing, saliva (1); Oesophagus: carries food (1); Stomach: digestion of proteins (1); Small intestine: digestion and absorption (1); Large intestine: water absorption (1). | 1+1+1+1+1 |
| | **OR:** Carbon dioxide and water (1); Sunlight (1); Chlorophyll in chloroplasts (1); Glucose and oxygen (1); In leaves (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

**Full Syllabus (Chapters 1–11)** | Class 7 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** a source of light, an opaque object and a screen | All three. |
| 2 | **(d)** 21 June | Summer solstice. |
| 3 | **(d)** base | Bases turn red litmus blue. |
| 4 | **(a)** hot | Heating effect of current. |
| 5 | **(d)** metal oxide | E.g. magnesium oxide. |
| 6 | **(a)** air and moisture | Both oxygen and water. |
| 7 | **(d)** growth spurt | Sudden increase. |
| 8 | **(a)** more | They absorb more radiation. |
| 9 | **(b)** odometer | Shows kilometres covered. |
| 10 | **(c)** breathing | Intake of O2 and release of CO2. |
| 11 | **(c)** leguminous plants | They fix nitrogen. |
| 12 | **(b)** reflection | Reflection. |
| 13 | **(a)** 27 days | Orbital period. |
| 14 | **(c)** basic | Bases are bitter and slippery. |
| 15 | **(b)** battery | A battery is a combination of cells. |
| 16 | **(c)** malleability | E.g. aluminium foil. |
| 17 | **(d)** chemical change | New substances appear. |
| 18 | **(b)** 11 to 19 years | Teenage years. |
| 19 | **(c)** conduction, convection, radiation | Standard modes. |
| 20 | **(d)** 3600 seconds | 60 x 60. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Distance travelled per unit time (1); m/s (1). | 1+1 |
| 22 | Ingestion, digestion, absorption (1); Assimilation and egestion (1). | 1+1 |
| 23 | Making of food by plants using sunlight (1); Carbon dioxide + water gives glucose + oxygen (1). | 1+1 |
| 24 | Luminous objects emit light (1); Non-luminous objects only reflect it (1). | 1+1 |
| 25 | Rotation: spin on its axis (1); Revolution: motion around the Sun (1). | 1+1 |
| 26 | Baking soda (1); Lime water or soap (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Carbon dioxide + water (1); In sunlight with chlorophyll (1); gives glucose + oxygen (1). | 1+1+1 |
| 28 | Look through three aligned cards with holes (1); Light seen when holes are aligned (1); Not when one is shifted (1). | 1+1+1 |
| 29 | Earth rotates (1); Side facing Sun has day (1); Other side has night (1). | 1+1+1 |
| 30 | Some are corrosive (1); Some are poisonous (1); Use indicators instead (1). | 1+1+1 |
| 31 | Wire melts on excess current (1); Circuit breaks (1); Appliance saved (1). | 1+1+1 |
| | **OR:** Malleable (1); Good heat conductor (1); Does not corrode easily (1). | 1+1+1 |
| 32 | Mix them in a bottle (1); Gas formed (1); Pass through lime water (1). | 1+1+1 |
| | **OR:** Secretes adrenaline (1); Helps in emergencies (1); Increases heartbeat (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers

**6 × 5 = 30**

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| **33** | Source, object, screen (1); Shadow forms behind (1); Moving the object closer to source enlarges (1); Moving closer to screen shrinks (1); Light travels straight (1). | 1+1+1+1+1 |
| **34** | Rotation: day and night (1); 24 hours (1); Revolution: year (1); 365.25 days (1); Seasons with tilt (1). | 1+1+1+1+1 |
| **35** | Citric: lemon (1); Acetic: vinegar (1); Lactic: curd (1); Tartaric: tamarind (1); Ascorbic: amla (1). | 1+1+1+1+1 |
| **36** | Battery is two or more cells (1); Join + of one to - of next (1); 1.5 x 3 = 4.5 V (1); More voltage (1); Brighter bulb (1). | 1+1+1+1+1 |
| | **OR:** Mercury liquid (1); Sodium soft (1); Graphite conducts (1); Iodine lustrous (1); Diamond hard (1). | 1+1+1+1+1 |
| **37** | Colour change (1); Gas evolution (1); Heat and light (1); Precipitate (1); Smell change (1). | 1+1+1+1+1 |
| | **OR:** Growth hormone (1); Thyroxine: goitre (1); Insulin: diabetes (1); Imbalance causes problems (1); Medical help (1). | 1+1+1+1+1 |
| **38** | Medium needed (1); Direction (1); Particle movement (1); Example (1); Speed (1). | 1+1+1+1+1 |
| | **OR:** Oscillation: one to-and-fro (1); Time period: time for one (1); Amplitude: extreme displacement (1); Bob returns (1); Periodic motion (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

**Full Syllabus (Chapters 1–11)** | Class 7 Science | Maximum Marks: 80

### SECTION — A (Objective Type) — Answers — 20 × 1 = 20

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** gills | Dissolved oxygen. |
| 2 | **(b)** make food for all | Base of the food chain. |
| 3 | **(b)** virtual and erect | Cannot be on a screen. |
| 4 | **(c)** Moon | Moon blocks Sun. |
| 5 | **(d)** $H_2SO_4$ | It contains two hydrogen atoms. |
| 6 | **(d)** insulators | E.g. rubber. |
| 7 | **(c)** oxygen | Oxygen is a gas. |
| 8 | **(d)** ripening of fruit | Cannot be undone. |
| 9 | **(b)** voice box (larynx) | Larynx enlarges. |
| 10 | **(a)** moving particles of the fluid | Moving fluid. |
| 11 | **(a)** metre per second | m/s. |
| 12 | **(b)** digestion | Digestion. |
| 13 | **(c)** photosynthesis | Photo means light. |
| 14 | **(d)** luminous objects | E.g. the Sun. |
| 15 | **(b)** 24 hours | It causes day and night. |
| 16 | **(c)** do not change the colour of litmus | They are neither acidic nor basic. |
| 17 | **(b)** electric iron | Heating element. |
| 18 | **(b)** basic | They turn red litmus blue in water. |
| 19 | **(c)** hydrated iron oxide | Reddish-brown flakes. |
| 20 | **(a)** girls | Girls begin earlier. |

### SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 21 | Connect positive of one to negative of the next (1); Cells joined in series (1). | 1+1 |
| 22 | Mercury (1); Bromine (1). | 1+1 |
| 23 | Presence of oxygen (1); Presence of moisture (1). | 1+1 |
| 24 | Monthly discharge of blood from the uterus (1); It occurs if the egg is not fertilised (1). | 1+1 |
| 25 | They reflect more heat (1); Keep us cool (1). | 1+1 |
| 26 | 400/80 (1); 5 m/s (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 6 × 3 = 18

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 27 | Hot air rises (1); Cool air sinks (1); Convection spreads it (1). | 1+1+1 |
| 28 | 72 x 5/18 = 20 m/s (1); 15 x 18/5 (1); 54 km/h (1). | 1+1+1 |
| 29 | Heart contracts and relaxes (1); Blood pushed into arteries (1); Veins return blood (1). | 1+1+1 |
| 30 | Leaves form a pitcher (1); Insects fall in (1); Enzymes digest them (1). | 1+1+1 |
| 31 | Image 2 m behind (1); Total 4 m (1); Equal distances (1). | 1+1+1 |
| | **OR:** Earth between Sun and Moon (1); Earth's shadow on the Moon (1); At full moon (1). | 1+1+1 |
| 32 | Bacteria make acid from sugar (1); Acid damages enamel (1); Toothpaste is basic (1). | 1+1+1 |
| | **OR:** Voltages add (1); Bulb glows brighter (1); Battery formed (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 6 × 5 = 30

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|

| | | |
|---|---|---|
| **33** | Kidneys filter blood (1); Urine in ureters (1); Stored in bladder (1); Passed through urethra (1); Waste removal (1). | 1+1+1+1+1 |
| **34** | Take in oxygen (1); Give out carbon dioxide (1); Roots respire in soil air (1); Shed leaves (1); Store gums, resins (1). | 1+1+1+1+1 |
| **35** | Three mirror strips in a triangle (1); Coloured pieces at one end (1); Multiple reflections (1); Symmetric patterns (1); Rotating changes patterns (1). | 1+1+1+1+1 |
| **36** | No atmosphere (1); Craters (1); Weaker gravity (1); Extreme temperatures (1); No water seen on surface (1). | 1+1+1+1+1 |
| | **OR:** Acid + base reaction (1); $\mathrm{HCl + NaOH \longrightarrow NaCl + H_2O}$ (1); Salt and water formed (1); Treating acidity (1); Treating acidic soil (1). | 1+1+1+1+1 |
| **37** | Cell: gives energy (1); Wires: path (1); Switch: control (1); Bulb: glows (1); Closed circuit needed (1). | 1+1+1+1+1 |
| | **OR:** Non-metals brittle (1); Dull (1); Poor conductors (1); Not sonorous (1); Metals are opposite (1). | 1+1+1+1+1 |
| **38** | Iron + oxygen + water gives rust (1); Needs air and moisture (1); Paint (1); Galvanise (1); Alloy as stainless steel (1). | 1+1+1+1+1 |
| | **OR:** Pituitary: growth hormone (1); Thyroid: thyroxine (1); Pancreas: insulin (1); Adrenal: adrenaline (1); Testes and ovaries: sex hormones (1). | 1+1+1+1+1 |

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